

M.TECH DISSERTATION PHASE-I • PROGRESS EVALUATION

Empirical Benchmarking of Neural Text-to-Speech Architectures for Indic Languages

Does explicit Devanagari grapheme-to-phoneme conversion still matter?

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nəmək → nəmkiːn



the vowel nobody writes

Hindi deletes many of the unwritten vowels Devanagari implies, and keeps others. The spelling does not say which. This study measures whether a neural TTS model still needs an explicit grapheme-to-phoneme front end to get that right, or learns it from audio.

- Three architectures from three generative families: FastSpeech 2, VITS, Matcha-TTS
- Two input arms, phonemic and graphemic, under a provably identical training budget
- A nested data ladder from 9 hours down to 10 minutes
- Marathi as a same-script control: same script, no medial schwa deletion
- This report covers the preparatory phase, which is complete. Nothing is trained yet.

नमक

namak (salt)

written n m k

segmented nə mə kə

final rule nə mə k(ə)

VC_CV nə mə k

/nəmək/

नमकीन

namkeen (savoury)

written n m kī n

segmented nə mə kī n

VC_CV nə m(ə) kī n

VC_CV nə m kī n

/nəmkī:n/

The first three letters are identical. The pronunciation of the second one is not, and nothing in the spelling says so.

Marathi is written in the same script and shares almost all of Hindi's phone inventory, but does not apply the medial rule. That is what makes it a control rather than a second target language.

Two words, one problem

- Devanagari is an abugida: every consonant carries an unwritten schwa
- The rule since Narasimhan, Sproat and Kiraz (2004): delete a schwa in V C _ C V, right to left
- Accurate, not perfect. Errors cluster at morpheme boundaries and in loanwords

2 बी G2P or not 2 बी?

Purists build the full front end. Modernists feed raw Devanagari in and assume the model learns zero duration schwas by itself. Nobody has umpired the argument.

PROBLEM STATEMENT

Holding training budget, evaluation protocol and speaker fixed, does phonemic input from an explicit Devanagari G2P front end improve neural TTS quality in Hindi relative to raw grapheme input?

Does it depend on the architecture?

FastSpeech 2 · VITS · Matcha-TTS

Does it survive less data?

9 h → 5 h → 1 h → 30 min → 10 min

Is it schwa, or is it script?

Marathi control: same script, no medial rule

OBJECTIVES

01

Speaker-matched corpus,
frozen checksummed splits

02

G2P front end with schwa
deletion as a switchable stage

03

Train all three under one
provably identical budget

04

Ladder, Marathi control,
verified metrics, release



Literature review: the architectures

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One from each current generative family, so the result is not a quirk of one design

FastSpeech 2

~27M

Non-autoregressive transformer

Generation

one parallel forward pass

Durations

predicted, needs a target

Vocoder

separate HiFi-GAN

Weakness

averaged, flat prosody

VITS

~83M

VAE + normalising flow + GAN

Generation

text straight to waveform

Durations

stochastic predictor

Vocoder

built in, none shared

Weakness

largest, 16 kHz checkpoint

Matcha-TTS

~18M

Optimal-transport flow matching

Generation

a few ODE solver steps

Durations

predicted, MAS aligned

Vocoder

separate HiFi-GAN

Weakness

least validated on Indic

StyleTTS 2 reports higher naturalness than any of these. It is cut for compute, not for quality, and is named as future work.

⌘ Literature review: the gap

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External forced alignment

A separate GMM-HMM aligner marks phone boundaries. Accurate, and what the original FastSpeech 2 recipe assumed.

Monotonic alignment search

Dynamic programming inside the model, improving as the model does. Used by VITS and Matcha-TTS.

Internal unsupervised aligner

Soft attention learned jointly, then sharpened to a hard path. Used here for FastSpeech 2.

Why forced alignment was dropped

Forced alignment would give FastSpeech 2 supervision the other two never get, and MFA has a Hindi model but no Marathi one. Both asymmetries would sit inside the comparison.

What no existing work does

Narasimhan et al.	rule only, no TTS
Arora et al.	G2P accuracy only
AI4Bharat	single system
IndicVoices-R	corpus contribution
MMS	one model, 1,000+ languages

None isolates the front end, none uses a same-script control, and none reports how the effect changes with data quantity.

Methodology: one variable at a time

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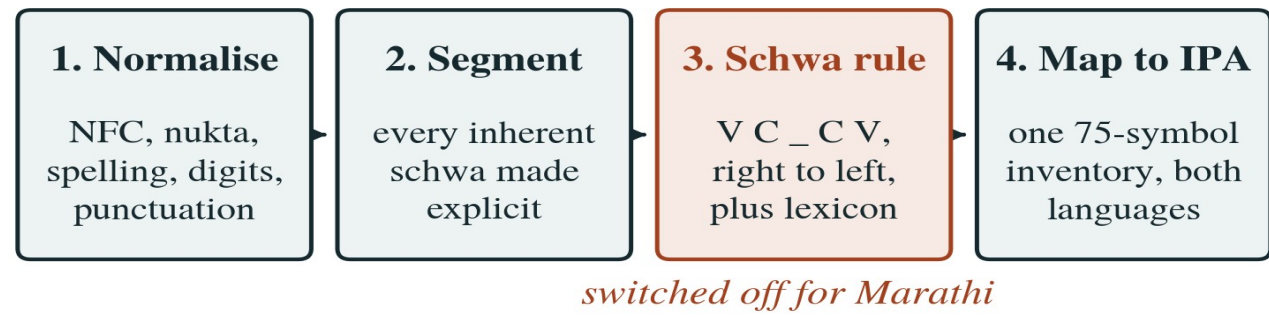
Held constant

- 100,000 steps, 12,000 mel frames/batch, one LR schedule, fp16
- One male speaker per language, matched on median F₀
- One HiFi-GAN per language, shared
- Same metrics, same 300-utterance test set

Varied

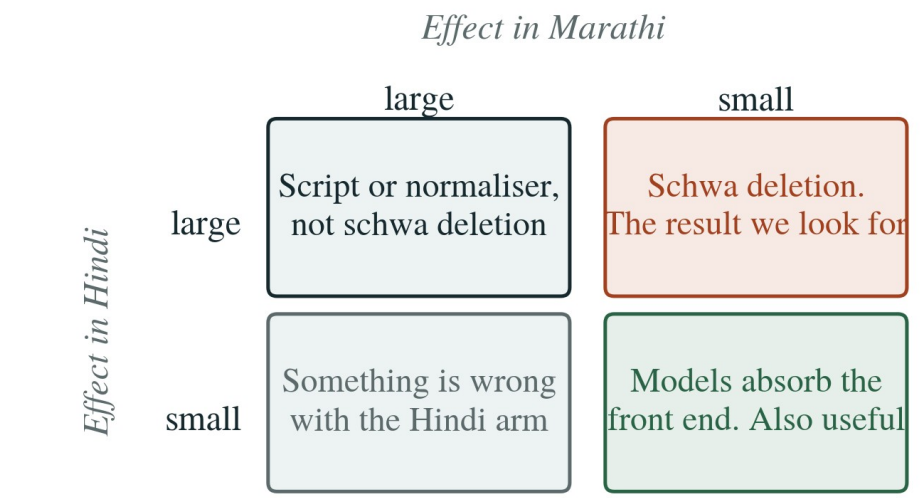
- Input representation: phonemes or graphemes
- Architecture: FastSpeech 2, VITS, Matcha-TTS
- Data quantity: five nested rungs
- Language: Hindi, Marathi as control

Methodology: the front end and the control



Stage 3 is the entire control condition

Marathi runs identical code with medial deletion disabled. One boolean flag. There is no separate Marathi code path, so the control cannot be contaminated by an implementation difference between the two languages. The grapheme arm runs stages 1 and 4 only, so both arms share one normaliser and the ablation measures phonology rather than preprocessing.



Why Marathi is the control

Same script, same corpus collection, same resource level. It differs from Hindi in the phonological rule and in nothing else that matters, which is what a control has to do.

Splits are frozen by a salted SHA-256 of the utterance id, so a rebuild reproduces the test set byte for byte, and the ladder takes prefixes of one ordering.



Results: the apparatus, measured

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Nothing is trained yet. Everything here is a property of the corpus, the front end or the metrics.

48.6 h

profiled across both
corpora, 22,764 utts

0

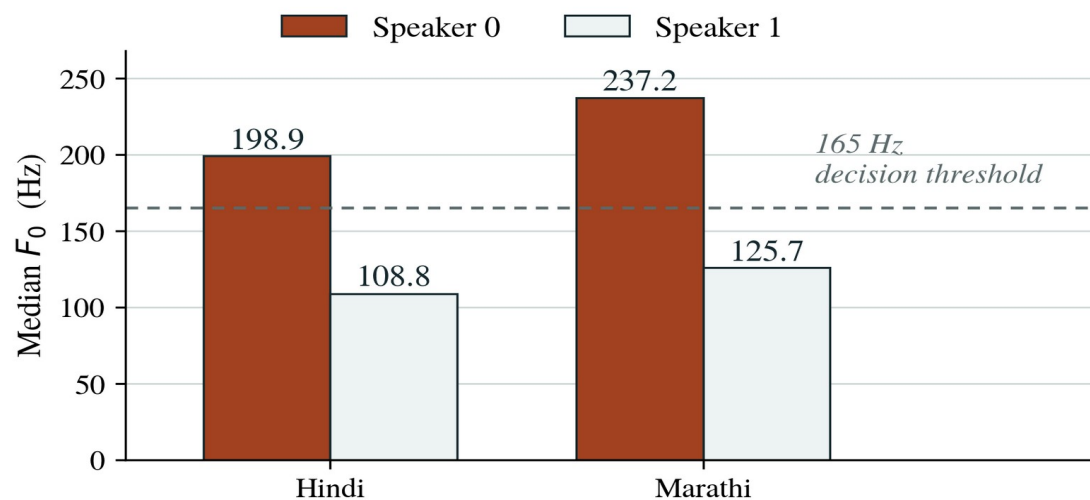
unknown symbols over
870,547 phone tokens

19%

error in the estimate
from a 50-row sample

78

tests passing, metrics
verified on paper



Speaker identity had to be recovered

The corpora label speakers 0 and 1 with no key, and the two arms must be matched. Median F₀ over 25 sampled utterances separates them cleanly and identically in both languages: 90 Hz apart in Hindi, 111 Hz in Marathi. Every gender label in the report rests on that measurement and is reported as an inference, not as something the dataset says.

Male speaker chosen in both: the two male voices sit 17 Hz apart across languages against 38 Hz for the female pair, so less acoustic difference is left that has nothing to do with schwa deletion.

ॐ Discussion: three defects, none of which announced itself

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	Correct	With a comma attached
input	नमक	नमक,
word-final schwa	deletes	survives
its left neighbour	kept	deletes
output	/nəmək/	/nəmkə/

Punctuation was rewriting words

A trailing comma means the word-final schwa is no longer final, so it survives, and its survival supplies the vowel that lets the schwa to its left delete. 16.5% of Hindi word tokens carry punctuation, and the damage fell on the phonemic arm alone.

A run was in the plan twice

The ladder's 9 h rung was a second copy of the main run. Invisible because the run id sat inside the config hash, which makes two identical runs look distinct. 10 GPU-hours. 19 runs became 17.

The sheet leaked its answers

Its worked examples were themselves items from the 49-word test set, handing every participant three answers. Found only because a reader asked about one. A check now refuses such a sheet at generation time.

All three were found by running over real data or by watching somebody use an instrument. The 78 unit tests caught none of them.

Conclusion and future work

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Done

- Corpora profiled, standardised, speaker-matched
- Splits frozen, checksummed, byte-reproducible
- G2P front end: zero unknown symbols on both corpora
- MCD and log-F₀ verified against analytic answers
- 17 runs specified, budget enforced by assertion

Immediate work

- Four training wrappers (FS2, VITS, Matcha, HiFi-GAN)
- ASR-WER with its ground-truth floor, predicted MOS, RTF
- Listening test infrastructure, piloted before recruiting
- Elicitation sheets from 2–3 native speakers (blocked on people)
- Kaggle GPU round trip, including checkpoint resume

Whichever way the result goes it is useful. If the architectures absorb schwa deletion, nobody building an Indic TTS system needs to write that front end again. If they do not, the field should stop defaulting to character level input for Indic languages.

References

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- The report carries the full list of 30 references in IEEE style.*

Thank you

Questions welcome.

The report carries the full list of 30 references in IEEE style, with the derivations, the full experimental matrix and four appendices.